

# What's Gone Wrong With CYBERSECURITY And HOW TO PROTECT OURSELVES

In today's world, where technology is deeply ingrained in our daily lives, system security has become an increasingly crucial aspect of our digital existence. System security is the practice of protecting computer systems and networks from unauthorised access, theft, damage, or misuse. Let us try to understand these threats and see how we can deal with them



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**T**oday, we are living in a world with more than 20 billion connected devices. We call these devices smart because they can do smart jobs. They know what to do; they are intelligent enough to find their own algorithms to work on. On top of that they are able to run smart programs.

Unfortunately, these programs are not always as smart as we think them to be. They often contain bugs. We have various security models to counter these bugs and we know how to secure our systems in this industry, but even then we continue to face new challenges with security.

So, the question is, what went wrong with the Internet of Things (IoT) devices and why the security that we have today is not enough? The three problems that could explain this are connectivity, extensibility, and complexity.

**Connectivity.** Today, there are over 20 billion connected devices available, which is about times the population of the world. If one of these devices is compromised, there is a good chance of the rest of the connected devices to get compromised as well. The scary part is that the most of the important nodes of a critical infrastructure are connected to the same internet.

**Extensibility.** We are using the same code over and over again. For example, various Linux kernels use the same code to function. Then there are Android devices for which no new code is being written. One of the major drawbacks of reusing the same code over and over again is that the persisting security flaws get propagated along the way.

**Complexity.** Computer programs get complex over time. Windows 10, for instance, has over 50 million lines of code. Google offers two billion lines of code. More lines of code directly equate to more bugs and flaws. These bugs act as a gateway for attackers and hackers.

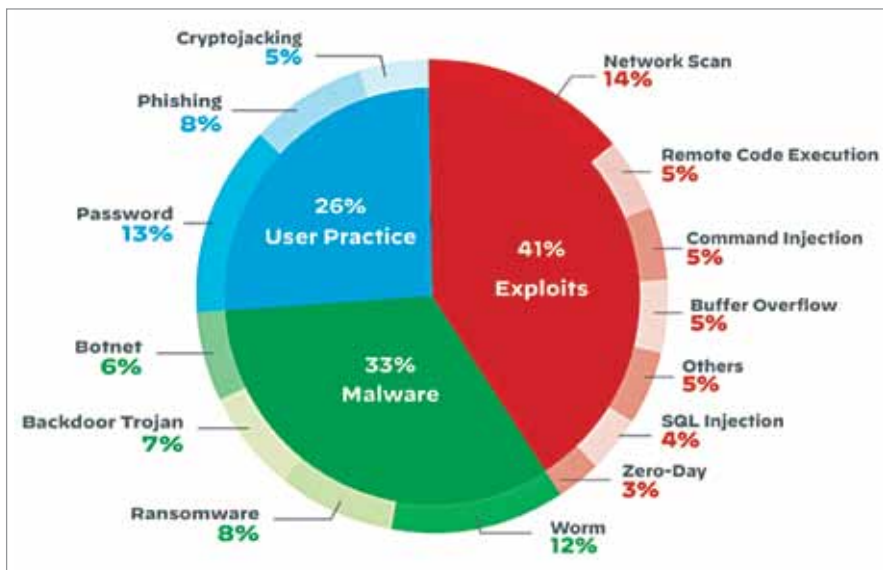


Fig. 1: Types of attacks with their total percentage (Source: Kaspersky)